

Exercise-induced lipid peroxidation: Implications for deoxyribonucleic acid damage and systemic free radical generation.



Exercise-induced deoxyribonucleic acid (DNA) damage is often associated with an increase in free radicals; however, there is a lack of evidence examining the two in parallel. This study tested the hypothesis that high-intensity exercise has the ability to produce free radicals that may be capable of causing DNA damage. Twelve apparently healthy male subjects (age: 23 ± 4 years; stature: 181 ± 8 cm; body mass: 80 ± 9 kg; and $\text{VO}_{2\text{max}}$: 49 ± 5 ml/kg/min) performed three 5 min consecutive and incremental stages (40, 70, and 100% of $\text{VO}_{2\text{max}}$) of aerobic exercise with a 15-min period separating each stage. Blood was drawn after each bout of exercise for the determination of ex vivo free radicals, DNA damage, protein carbonyls, lipid hydroperoxide (LOOH) concentration, and a range of lipid-soluble antioxidants. Lipid-derived oxygen-centered free radicals (hyperfine coupling constants $a(\text{Nitrogen}) = 13.7$ Gauss (G) and $a\beta(\text{Hydrogen}) = 1.8$ G) increased as a result of acute moderate and high-intensity exercise ($P < 0.05$), while DNA damage was also increased ($P < 0.05$). Systemic changes were observed in LOOH and for lipid-soluble antioxidants throughout exercise ($P < 0.05$); however, there was no observed change in protein carbonyl concentration ($P > 0.05$). These findings identify lipid-derived free radical species as possible contributors to peripheral mononuclear cell DNA damage in the human exercising model. This damage occurs in the presence of lipid oxidation but in the absence of any change to protein carbonyl concentration. The significance of these findings may have relevance in terms of immune function, the aging process, and the pathology of carcinogenesis. Copyright © 2010 Wiley-Liss, Inc.